

Localization of Curving Beams

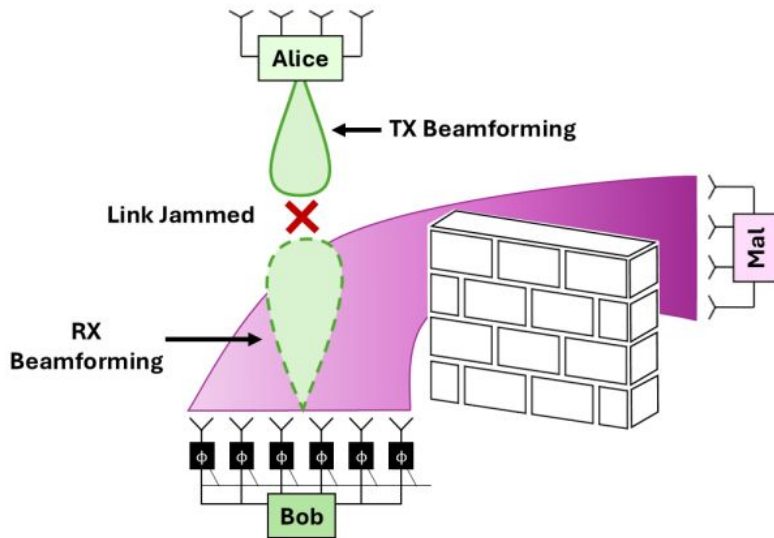
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Problem Statement

Can we localize a curving beam source

- Prior work utilizes 2D “heatmap” measurements to characterize the trajectory of beams
- 2D measurements are impractical for real use cases

Is it possible to estimate TX aperture parameters using only a 1D RX aperture ?

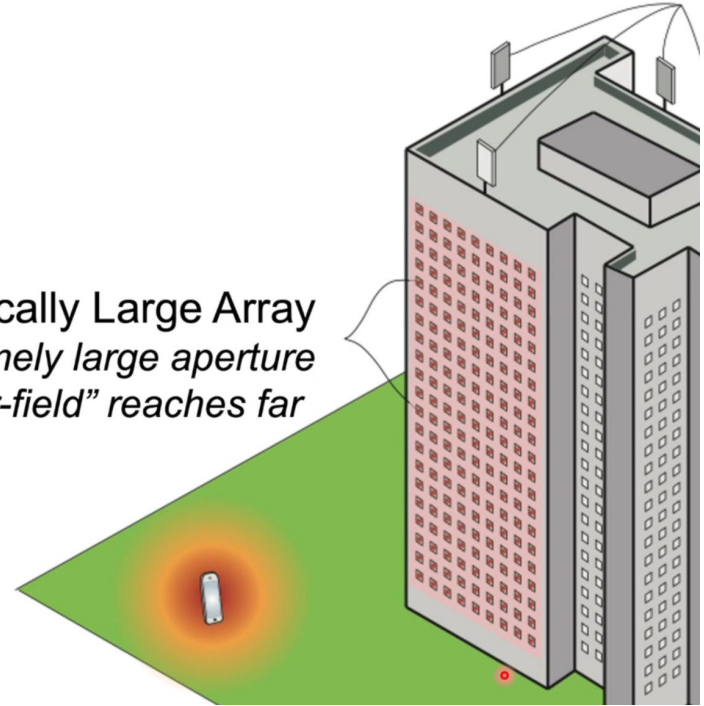


(c) Layer 3: Main lobe jamming.

Jamming and Communications application

- Beam parameter estimation is valuable
- Beam parameters can be changed by objects in the environment.
- This work is motivated by more systems being developed where users are in the near-field

Physically Large Array
Extremely large aperture
"Near-field" reaches far



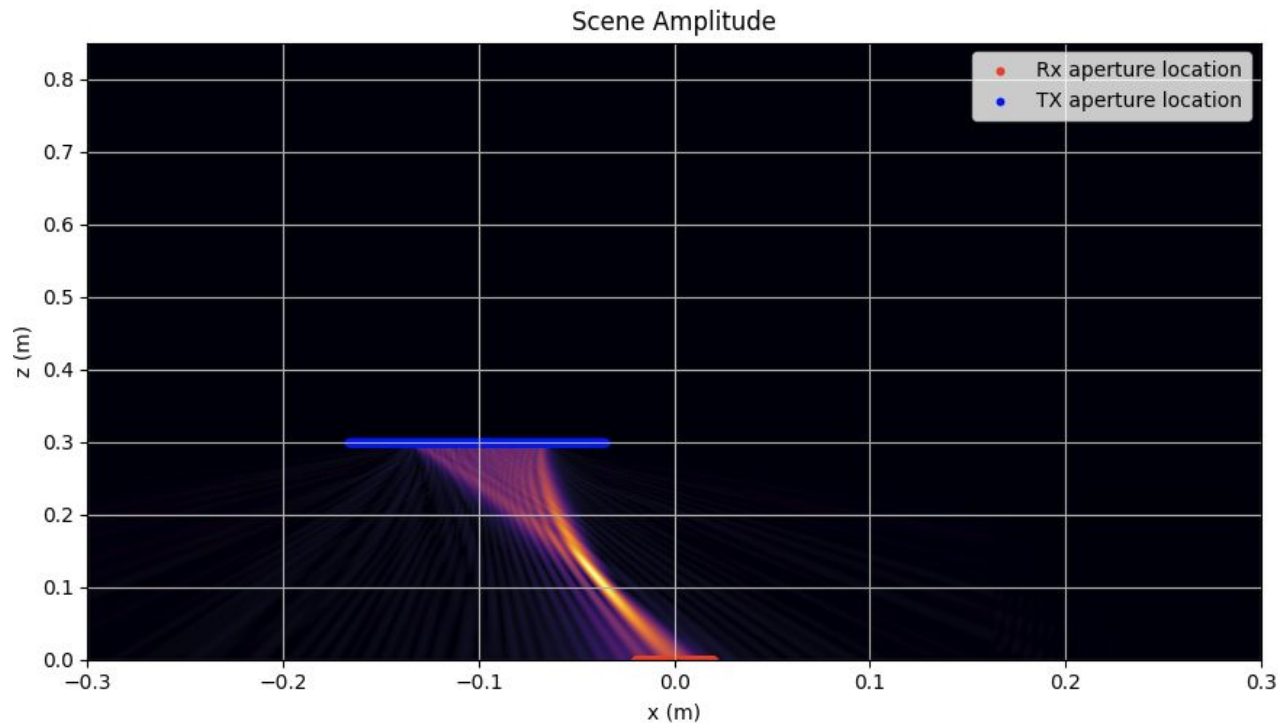
Phase Retrieval Algorithms

- The aperture of a transmitter contains information about the beam it creates
- Phase retrieval algorithms can estimate a TX aperture
- Existing Algorithms such as Gerchberg-Saxton must be modified to work in this use case
 - Propagation model
 - Gradient Descent
 - Introduction of phase measurements
 - **Use of multiple frequencies**

Problem: Phase retrieval algorithms do not guarantee a unique solution, and require geometry of the TX aperture to be known

MGS example

When TX aperture location (blue) is already known, MGS can be used to estimate the TX aperture using only 1D RX measurements (red)



Proposed Solution

1. MGS paired with a simple grid search generates a set of candidate beams
2. Candidates are ranked by how well their reconstruction fits the RX measurements

Grid Search Implementation

Currently, candidate beams are selected by:

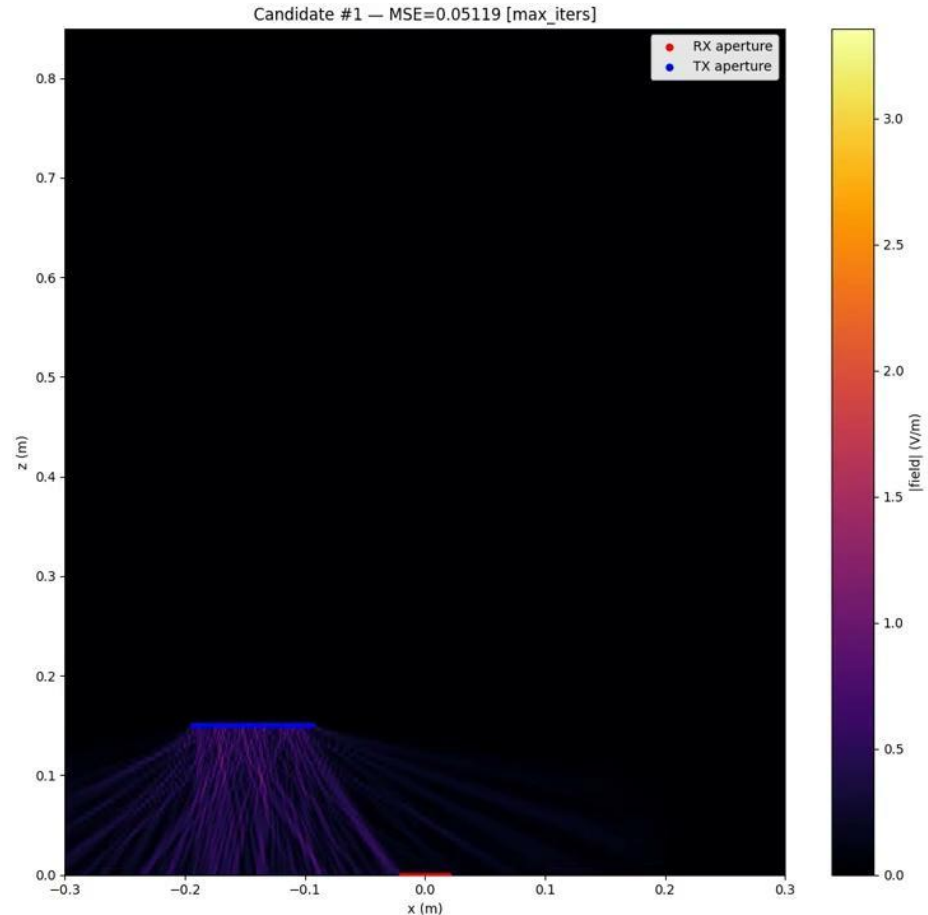
1. Picking a candidate TX aperture location
2. Running MGS using the known RX aperture measurements and candidate TX location
3. Wait for MGS to converge, then save data
4. Move on to to next TX aperture location and repeat

Grid Search Example

A candidate TX location is chosen, and MGS algorithm is run using real RX data.

Each MGS run generates a candidate TX aperture.

Some candidates are able to fit the RX measurements well, others are not.



Modifications of MGS

1. Propagation model (prior work)
2. Gradient Descent (prior work)
3. Introduction of phase measurements
4. **Inclusion of multiple frequencies directly into GS**
 - a. Supported by literature
 - b. Novel when used for this purpose with the other modifications in tandem
 - c. Matches how modern communication systems operate

Multi-frequency support for MGS

- Prior work ran MGS multiple times, and averaged the results
- Now multiple frequencies are solved for in a single run
- Serves to further constrain the problem and remove ambiguity
 - Each beam bends slightly differently at different frequencies
- **Introduces a lot more local minima that tricks gradient descent into incorrect solutions**

Warm-start for MGS with multiple frequencies

- To avoid getting stuck on a bad solution, MGS is run on 1 frequency first as a “warm start” and that solution is then iterated on using multiple frequencies
- This practice is common in the relevant literature

Ranking Success and Failure

- Candidate beams are ranked by residual error
- The candidates with the best residual error and physically close to each other are considered the best candidates

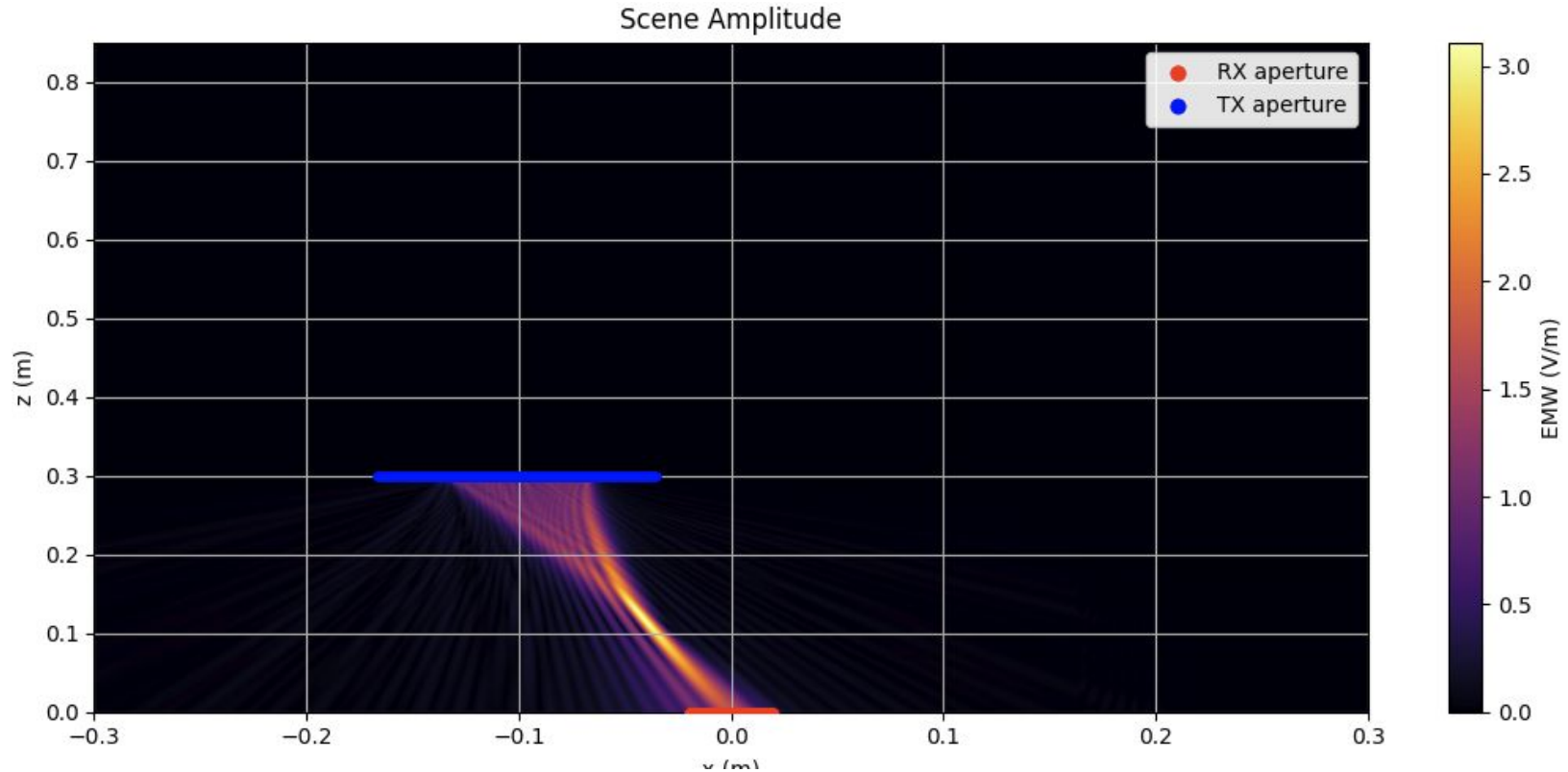
Scenarios Simulated and Tested

- $\lambda/20$ scenario with variable amounts frequency
- $\lambda/2$ scenario with variable amounts of bandwidth
- $\lambda/20$ vs $\lambda/2$ results

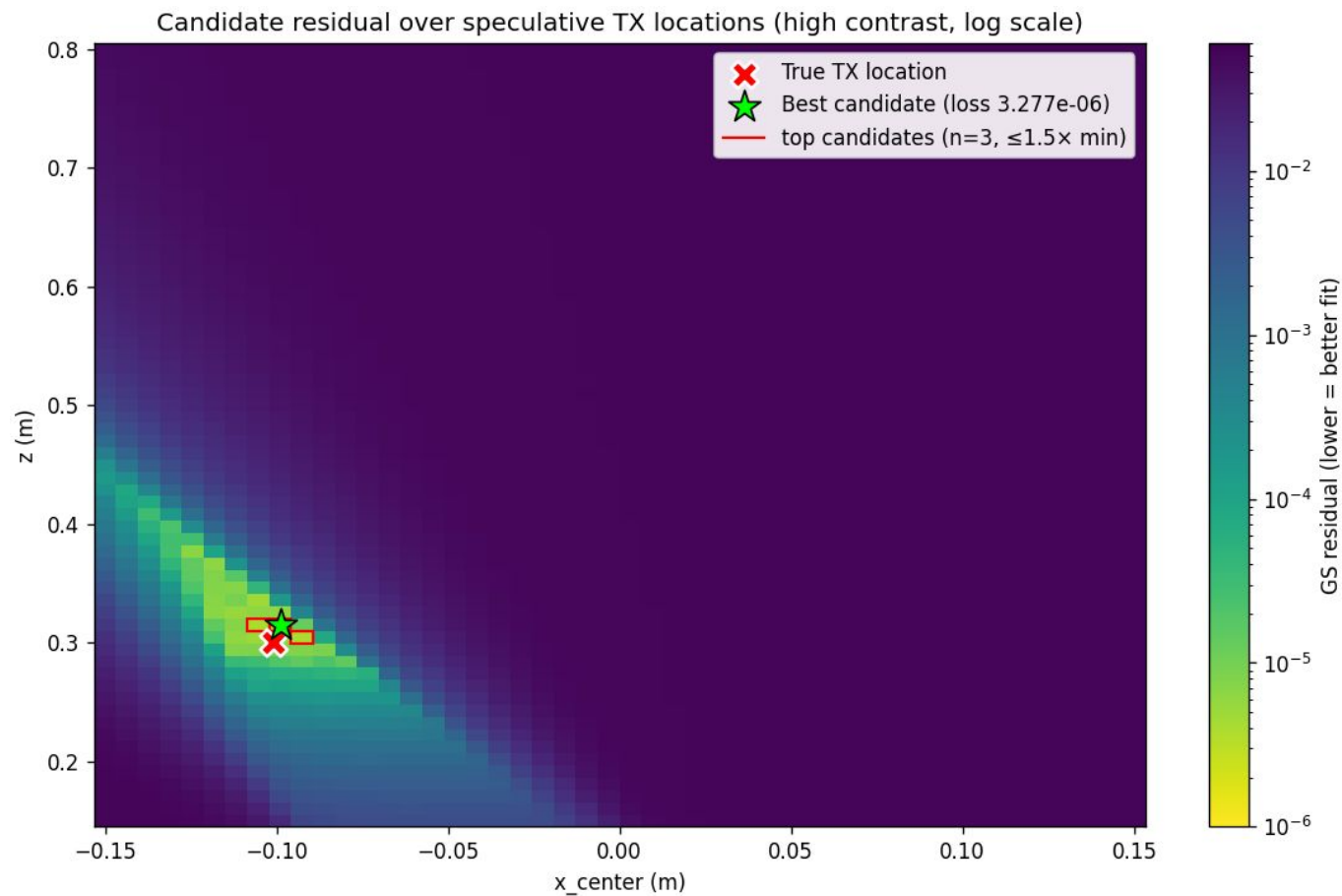
Best Case Scenario Simulation parameters

- $\lambda/20$ RX antenna element spacing
- Majority of energy is captured by RX antennas
- No noise, unrealistically high SNR
- Base of the beam is about 50mm
- 64x48 grid search. 3072 locations tested.
- Variable amount of bandwidth used
 - 1 freq
 - 1-20% spread

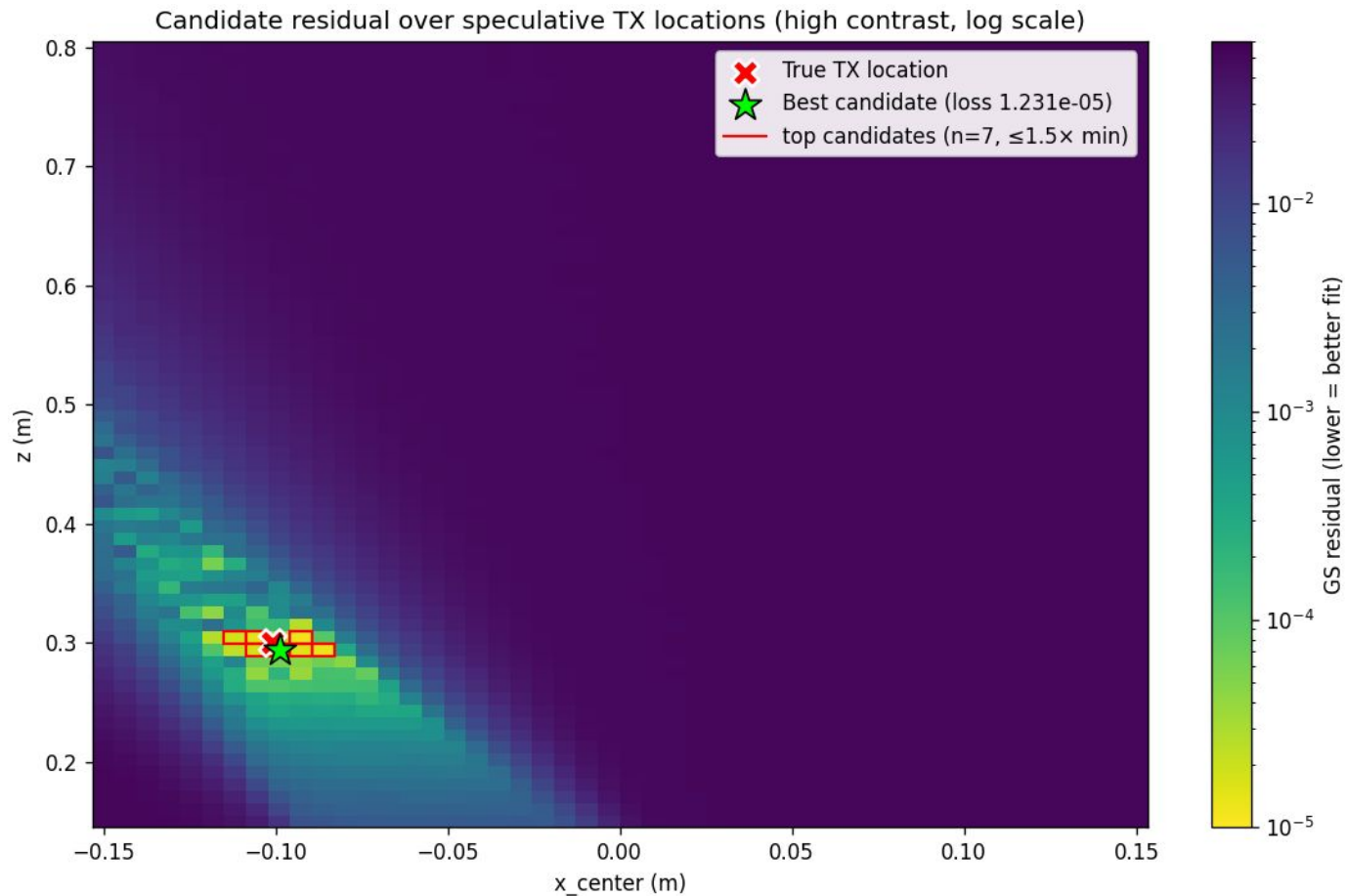
$\lambda/20$ Scenario True Beam and location



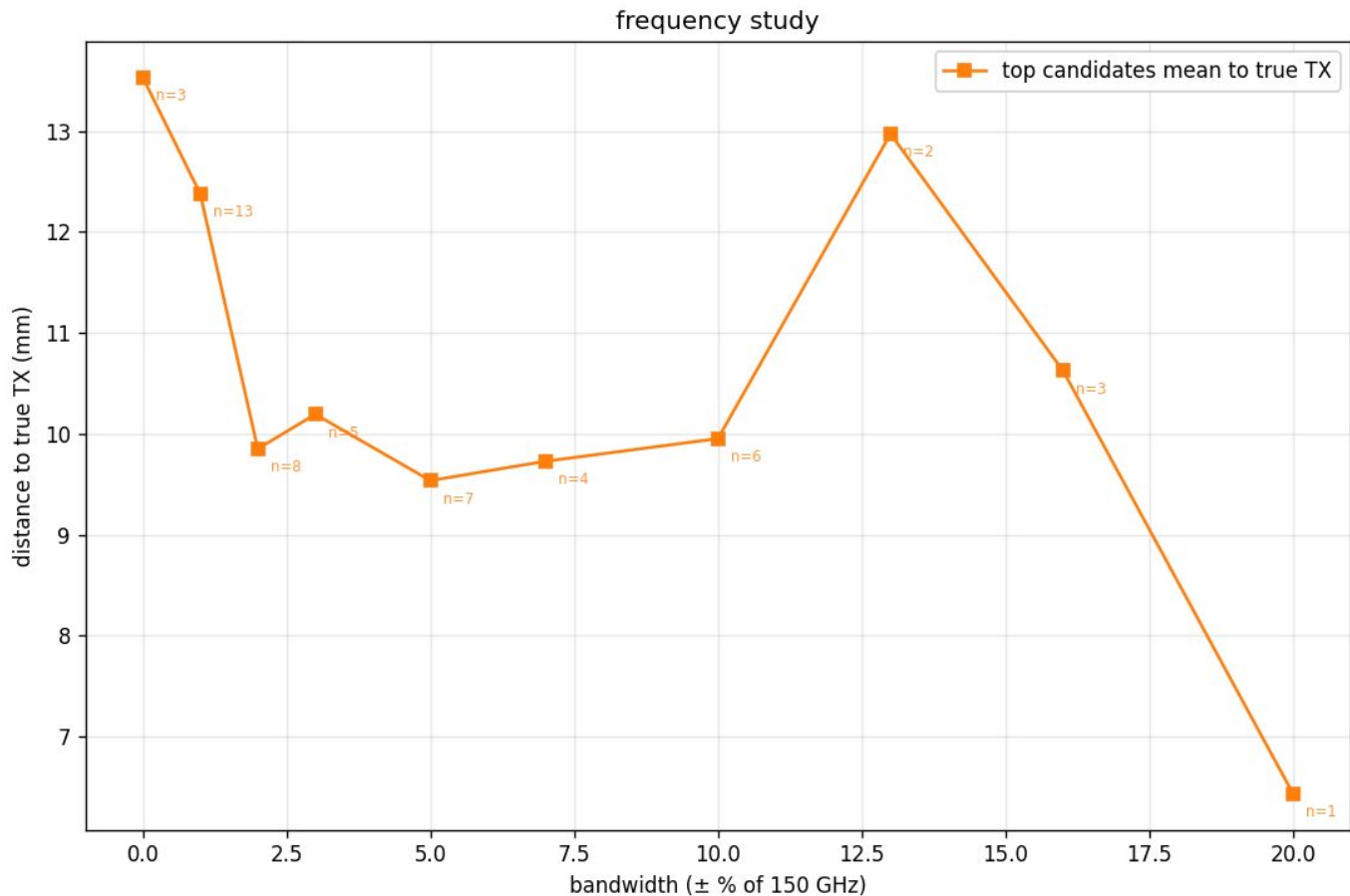
$\lambda/20$ Scenario Simulation results: 1F



$\lambda/20$ Scenario Simulation results: $\pm 10\%$



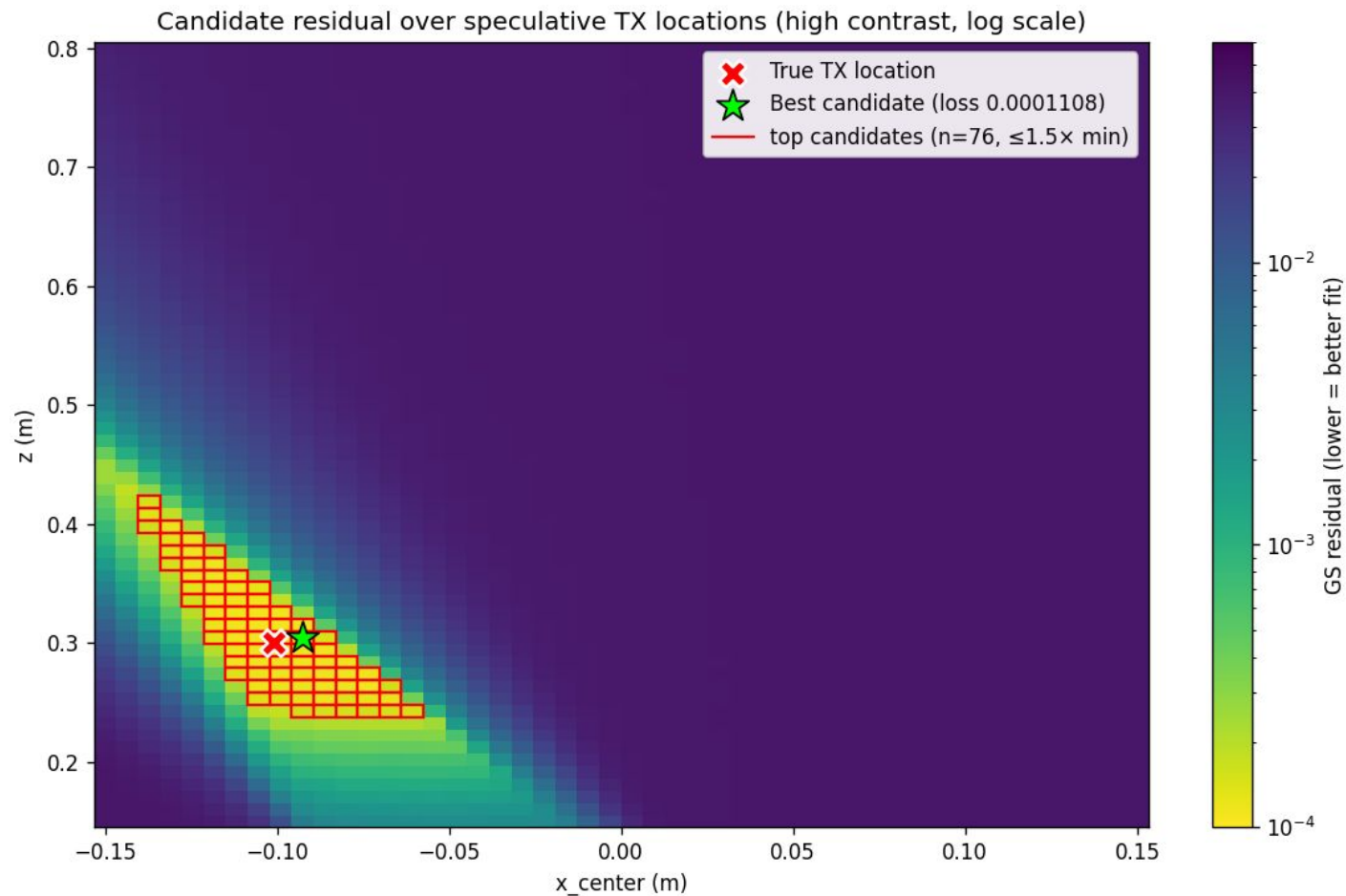
$\lambda/20$ Scenario: frequency vs distance error



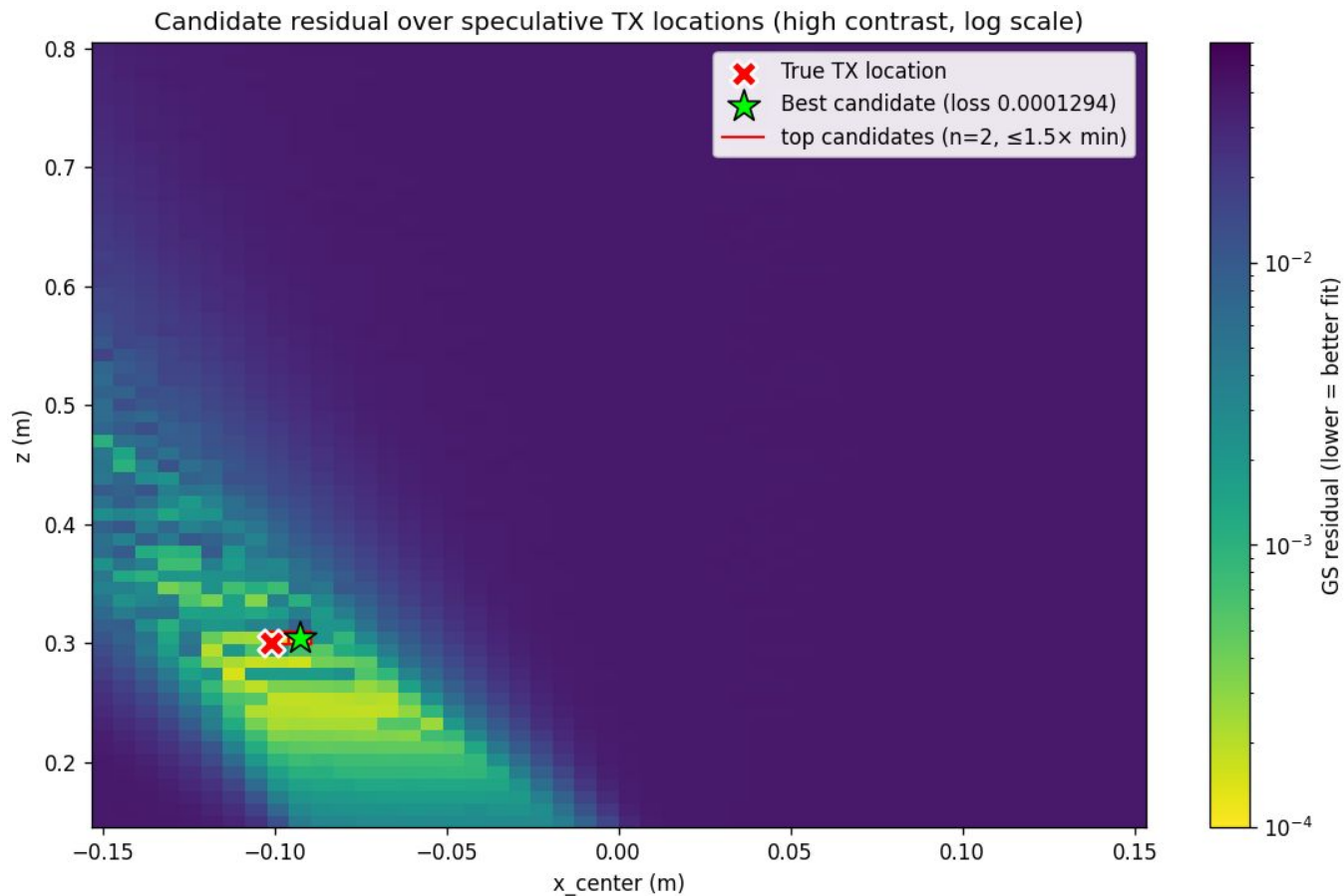
$\lambda/20$ Scenario Conclusion

- Using more does bandwidth improve results
- Tx aperture was $\sim 50\text{mm}$
- Using only 1 frequency got within 15 mm which is good enough
- In this case using bandwidth was not meaningful

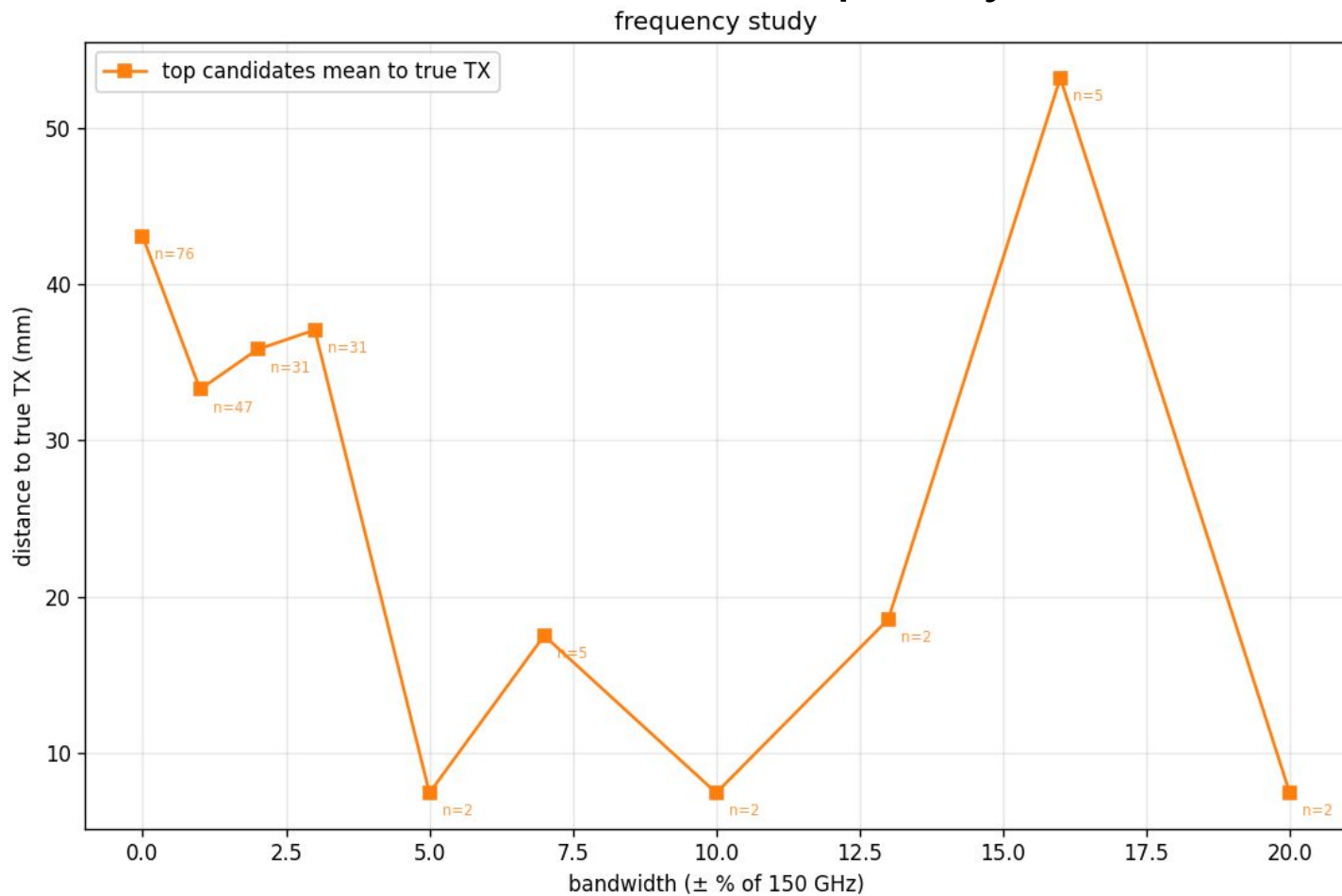
$\lambda/2$ Scenario Simulation results: 1F



$\lambda/2$ Scenario Simulation results: $\pm 10\%$



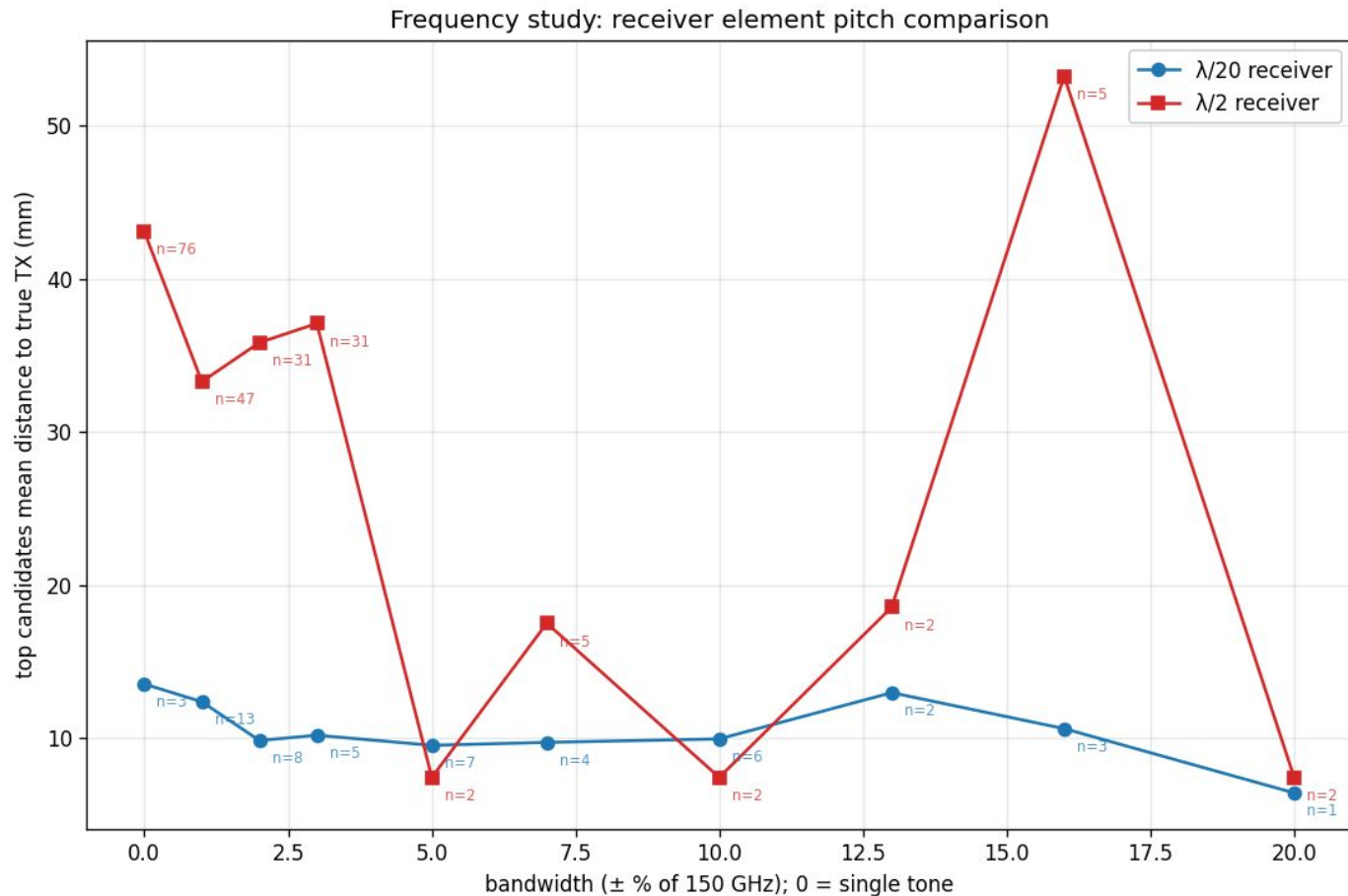
$\lambda/2$ Scenario Simulation results: frequency vs distance error



$\lambda/2$ Scenario Conclusion

- Reducing the sampling distance worsens performance meaningfully
- Tx aperture was $\sim 50\text{mm}$
- Using 5% or more bandwidth is required to get results below 15mm

$\lambda/20$ vs $\lambda/2$



Simulation Conclusions

- The proposed system works well in simulation
- Partial hits make it more difficult to define the boundaries of the region of interest
- Using more spectrum improves results
- With a artificially high SNR, the need for bandwidth is very low

Next Steps

- Test with real data OTA
- Quantify “information” or “degrees of freedom” somehow
 - Lowering SNR and lowering how much power is received by the RX aperture, and increasing sampling distance should all lower the amount of information, and therefore lower the results
 - TSVD method?
- Move on to next step of the problem:
 - Trajectory reconstruction
 - countermeasures

Questions

Supplemental Slides

Standard GS flowchart

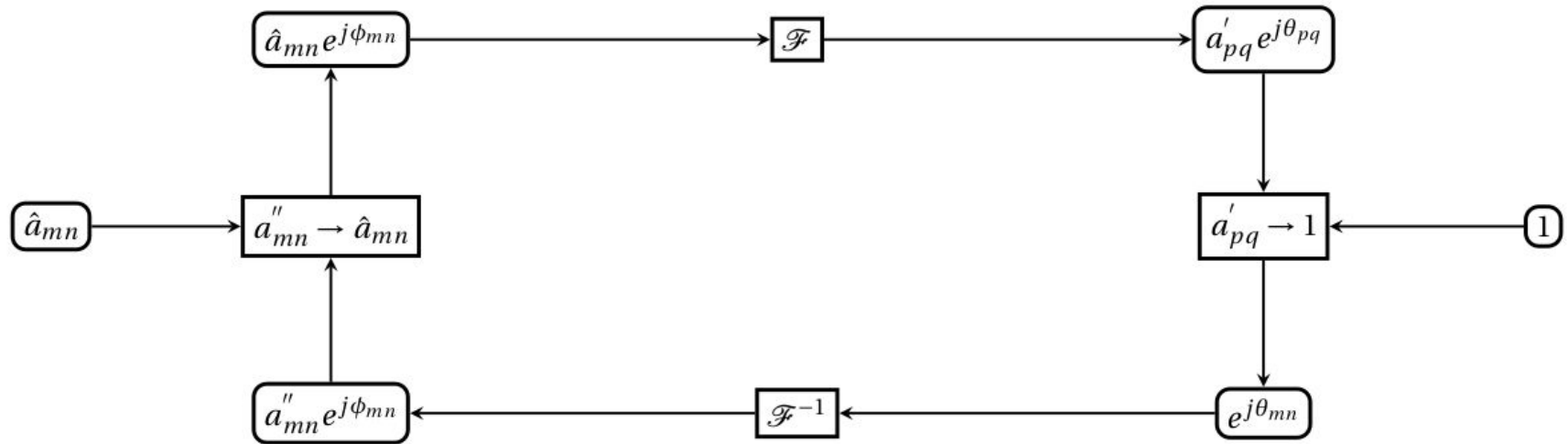


Figure 1. Flow chart for the Gerchberg-Saxton (GS) algorithm.

MGS with Gradient Descent - closest analog in literature

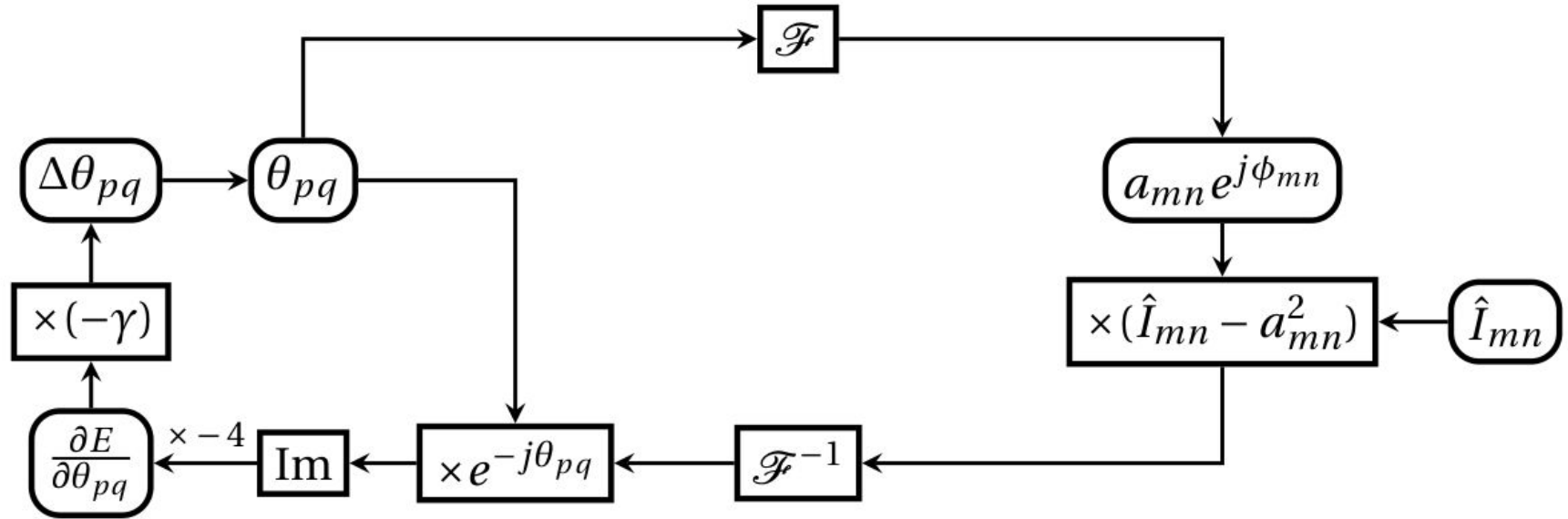


Figure 2. Calculation process of the proposed technique

MGS with Gradient Descent (new)

Testing in simulation showed that standard GS was unlikely to converge to the correct solution.

- Standard MGS iterates by simply forcing the Amplitude of the signal at the RX aperture to be what was measured and then allowing the phase to iterate
- This process is replaced by a error minimization algorithm
- This has been shown to work at optical frequencies but not at sub-THz
 - Shujian Liu and Yasuhiro Takaki. Optimization of phase-only computer-generated holograms based on the gradient descent method. Applied sciences, 10(12):4283–, 2020.
- Error is computed by comparing the Measured RX aperture to the computed RX aperture
- Error computation is propagated back to TX aperture, and gradient is computed
- Changes in the TX phase are made to minimize the error and process repeats

MGS with Phase Measurements

- Phase measurements at the receiver are required for more accurate error computations
- Gradient Descent algorithm is very sensitive to having an accurate error measurement
- Prior MGS gradient descent work only computed error using Amplitude differences

Phase Retrieval Using Gerchberg-Saxton

- Gerchberg-Saxton (GS) is an iterative phase retrieval algorithm
 - The Electric Field is propagated between the TX and RX aperture using the Fraunhofer approximation
 - Small changes are made each iteration until the algorithm converges on a solution
- **Multiple** modifications are required for this algorithm to work on sub-THz curving beams
 - Propagation model
 - Gradient Descent
 - Introduction of phase measurements
 - Use of multiple frequencies

MGS with Rayleigh-Sommerfeld

The propagation is replaced is replaced with Rayleigh-Sommerfeld

- Its Computationally more expensive but more accurate in the near field
- Proven to function as GS modification the sub-THz regime
 - Sudheer Jawla, Jean-Philippe Hogge, and Stefano Alberti. Theoretical investigation of iterative phase retrieval algorithm for quasi-optical millimeter-wave rf beams. IEEE Transactions on Plasma Science, 37(3):403–413, 2009.